INCH-POUND
MIL-PRF-1/1512C
2 December 2008
SUPERSEDING
MIL-E-1/1512B (USAF)
29 January 1971

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7248

Inactive for new design
after 5 December 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Tetrode
See figure 1
Mounting position: See note 2
Weight: 2.5 pounds nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Ebb	eb	Ec1	Ec2	ik	Pg1	Pg2
Unit:	V	kVdc	kV	Vdc	Vdc	a	W	W
Oil immersed	6.3 ±5%	125 (See note 1)	125	-600	1,500	2.0	10	20
TEST CONDITIONS:	6.3	----	----	----	1,000	----	----	----

ABSOLUTE RATINGS:

Parameter:	Pp	Glass temp	Insulation and cooling	T (coolant)
Unit:	W	°C	---	°C
Oil immersed	200	150	(minimum) See note 2	75 (See note 5)
Air insulation:	---	150	See note 3	----
Test Conditions :	----	----	See note 4	----

GENERAL:

Qualification – Not Required.

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TABLE I. Testing and inspection.

Inspection	MIL-STD-1311 Method	Conditions	Acceptance Level 12/	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1							
Positive grid drive	------	Ec1 = -150 Vdc; ec1 = 200 v; Ebb = 2,00 Vdc; tp = 160 µs (max) prr = 15 (max) (See note 6)	0.65	ic1 ic2 ib	--- --- 900	140 250 1,570	ma ma ma
High-voltage stability	-------	Ec2 = Ek =0; Ec1 = -500 Vdc; Ebb = 125 KVdc; Rp = 1.0 MegΩ (max) t = 30 min (max) (See note 7)	0.65	--	------	-----	----
Electrode voltage (grid)	1261	Ebb = 125 KVdc; Rp = 1.0 MegΩ (max) Ec1/∆Ib+20 µAdc (See note 8)	0.65	Ec1	-----	- 420	Vdc
Conformance inspection, part 2							
Filament current	1301	Ef = 6.3 V (See note 4)	------	If	10.0	13.5	A
Total grid current	1266	Ebb = 1,200 Vdc; Ec1/ib = 0.165 Adc; Ic2 = 0.02 Adc (max); t = 5 minutes (see notes 9 and 10)	-------	Ic1	----	- 50	µAdc
Quality conformance inspection, part 3							
Peak emission (See note 11)	1231	Ec1 = ec2 = eb = 400 V Ec1 = ec2 = eb = 2400 V	------	is is	1.5 5.0	3.2 ----	A A

NOTES:

1. When the tube is first installed, or after tube has been inoperative over a period of time, anode voltage, when first applied, shall be increased gradually from 75 kV to 125 kV, over a 10- minute period, in accordance with the manufacturer's recommendations. Warning: Operation of this tube may produce X-ray radiation. Adequate radiation protection shielding shall therefore be provided.

2. When immersed in oil horizontally, the tube will be cooled by natural convection. When the tube is operated vertically in oil, the re-entrant portion of the cathode end of the tube shall be cooled by forcing oil through the center pin No. 5 of the cathode base before and during the application of filament voltage at a minimum flow of 3 pin per minute. For oil insulation, the tube shall be immersed in a high-grade, low-viscosity insulation oil having a dielectric strength not less than 25,000 peak volts/0.1 inch when tested per ASTM test D-877.

3. When the tube is air tested and the tube is in any position, the re-entrant portion of the cathode end of the tube shall be cooled by forcing air thought the center pin No. 5 of the cathode base before and during application of filament voltage at a minimum flow rate of 1-1/2 cfm.

4. All test to be performed with the tube immersed horizontally in insulating oil as described in note 2. Filament current test may be done in air as described in note 3.

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5. Coolant temperature indicated is average temperature of oil surrounding tube and temperature of incoming oil to cathode when mounted vertically.

6. The positive pulse voltage, ec1, is measured between control grid and cathode. The amplitude of the voltage and current pulses shall be measured approximately at the middle of the pulse.

7. This test may be preceded by an aging period at reduced anode voltage of a duration not exceeding 10 minutes. The tube shall operated during the last 10-minute period with no more than two flashes. Total test time, excluding aging, not to exceed 30 minutes. Flashes shall be interpreted as luminescence within the vacuum space of the tube. Fluorescence of the glass surfaces of the tube shall not be considered a flash. Flashes may be counted by an electronic device, operation of which is based on the anode current surge which accompanies the flash.

8. Read Ib at Ec1 = -500 Vdc. Then make Ec1 less negative to the point where Ib increases by the indicated amount.

9. If impossible to obtain 0.165 ampere anode current without exceeding 0.020 ampere screen current, it shall be permissible to adjust anode voltage until such conditions are obtained provided no rating are exceeded.

10. To be performed prior to positive grid drive test.

11. The number of tubes to be tested shall be one percent of each month of each month's production, but not less than one nor more then three tubes shall be tested.

12. This specification sheet utilizes an accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

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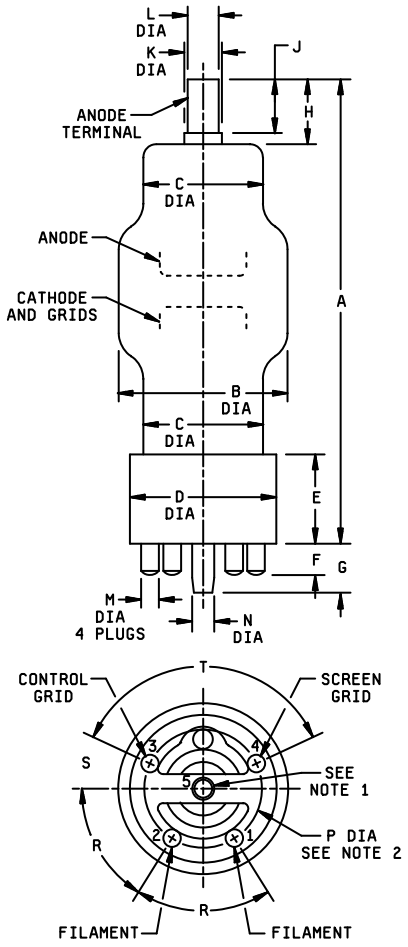


FIGURE 1. Outline drawing of electron tube type 7248.

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Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Conformance inspection, part 2		
A	9.562 (242.87)	9.938 (252.43)
B		3.625 (92.08)
C		2.625 (66.68)
D	3.00 (76.20)	3.063 (77.80)
E	1.812 (46.02)	1.936 (49.17)
F		.625 (15.88)
G	.969 (24.61)	1.031 (26.19)
H	1.188 (30.18)	1.438 (36.53)
J	1.063 (27.00)	1.125 (28.58)
K	.719 (18.26)	.781 (19.84)
L	.594 (15.09)	.656 (16.66)
M	.370 (9.40)	.380 (9.65)
N	.432 (10.97)	.442 (11.23)
P	2.490 (63.25)	2.510 (63.75)
R	58°	62°
S	28°	32°
T	118°	122°

NOTES:
1. Inlet tube for forced oil or air cooling (.156 inch (3.96 mm) I.D.).
2. Four pins located on circle diameter P.

FIGURE 1. Outline drawing of electron tube type 7248. Continued

Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311
ASTM-D877

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Air Force – 85
DLA- CC

Preparing activity:
DLA- CC
Project number (5960-2008-072)

Review activities:
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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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